

EXPERIMENT: 9

A PYTHON PROGRAM TO IMPLEMENT KNN AND K-MEANS.

Aim:

To implement K-Nearest Neighbors (KNN) and K-Means clustering.

Procedure:

KNN:

1. Import libraries.
2. Load dataset (Iris dataset).
3. Split into training and testing data.
4. Train KNN classifier.
5. Predict and evaluate accuracy.

K-Means:

1. Import libraries.
2. Load dataset.
3. Apply K-Means clustering.
4. Assign clusters to data points.
5. Visualize clusters.

CODE:

KNN

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score
```

```
data = load_iris()

X = data.data

y = data.target


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


model = KNeighborsClassifier(n_neighbors=3)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)


print("KNN Accuracy:", accuracy_score(y_test,
y_pred))
```

K-Means

```
from sklearn.datasets import load_iris

from sklearn.cluster import KMeans

import matplotlib.pyplot as plt


data = load_iris()


X = data.data

model = KMeans(n_clusters=3, random_state=42)

model.fit(X)


labels = model.labels_

plt.scatter(X[:, 0], X[:, 1], c=labels)

plt.xlabel("Feature 1")

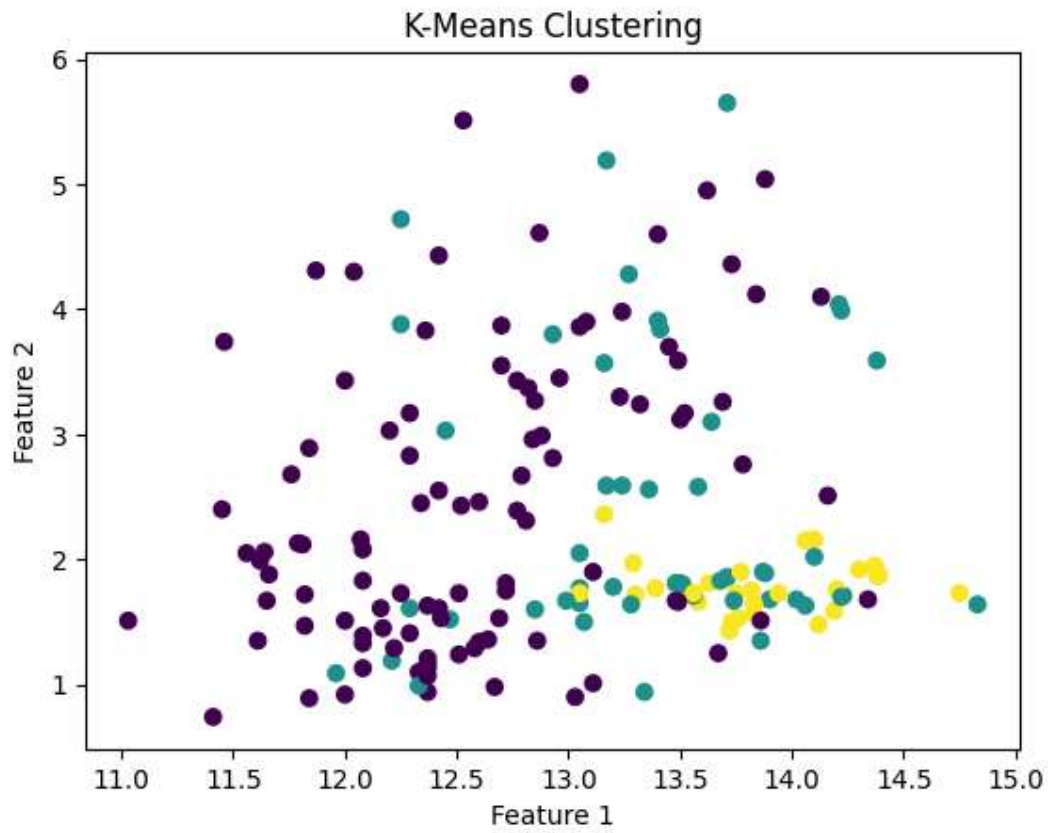
plt.ylabel("Feature 2")

plt.title("K-Means Clustering")

plt.show()
```

OUTPUT:

KNN Accuracy: 0.805



RESULT:

KNN successfully classified the data, and K-Means grouped the data into clusters effectively.